Topic Models as an Object Lesson in Machine Learning

Alexander Hoyle, 2025-03-02

Two sides of machine learning methods

Tools to solve problems

- Engineering-oriented
- Real-world use
- Extrinsic metrics, benchmarks

Vessels for ideas

- Preference for elegance/simplicity
- Theory-driven
- Intrinsic metrics

Two sides of machine learning methods

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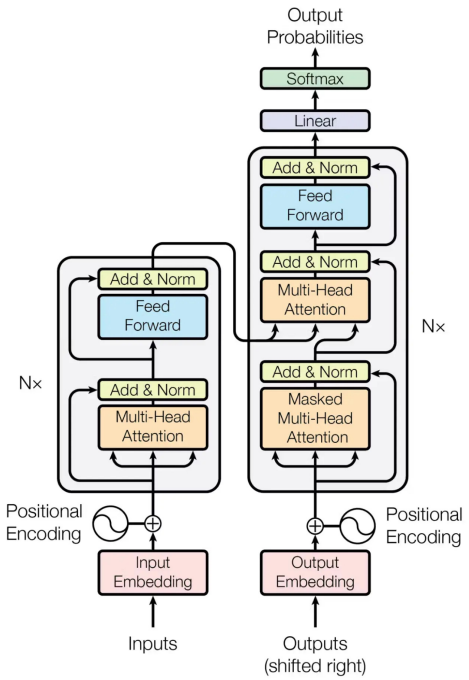

Vessels for ideas

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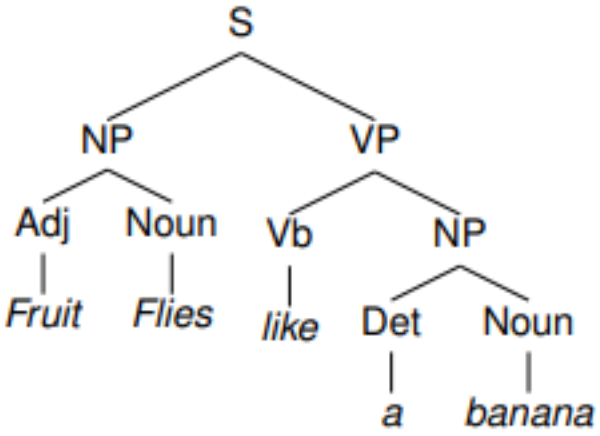
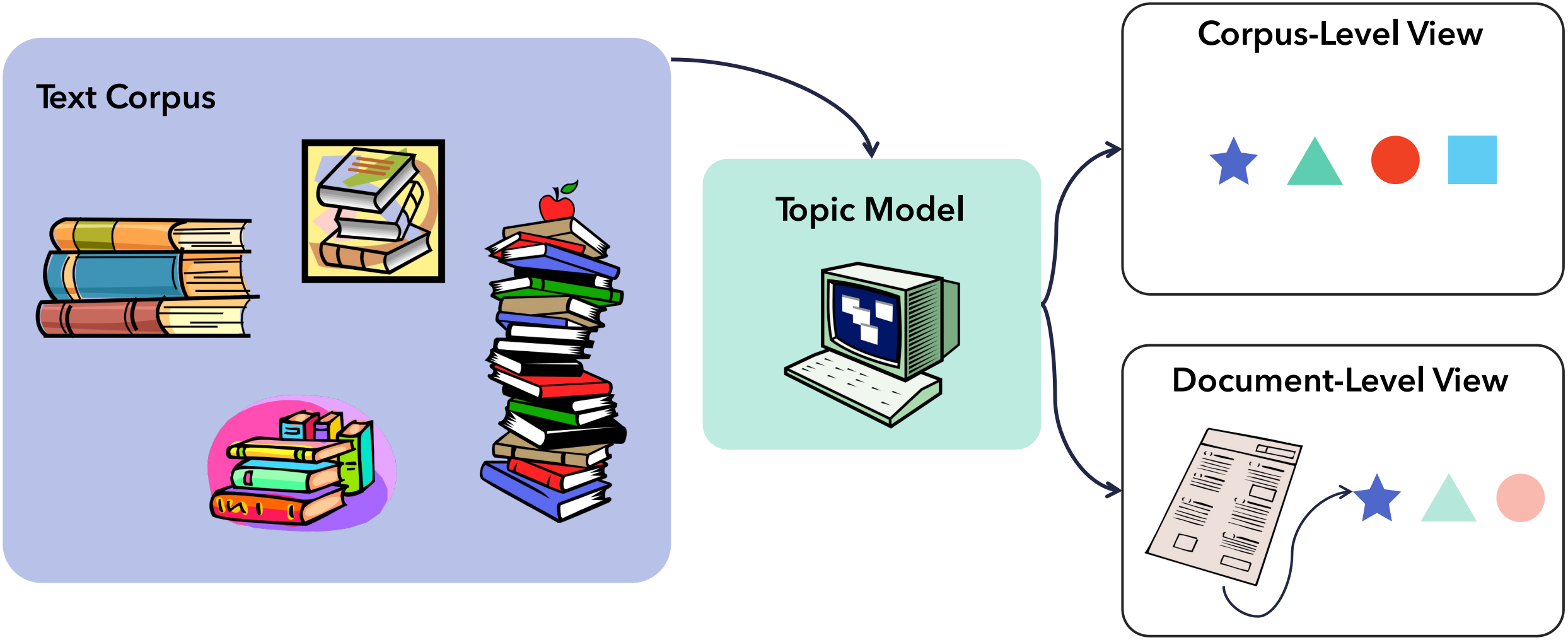
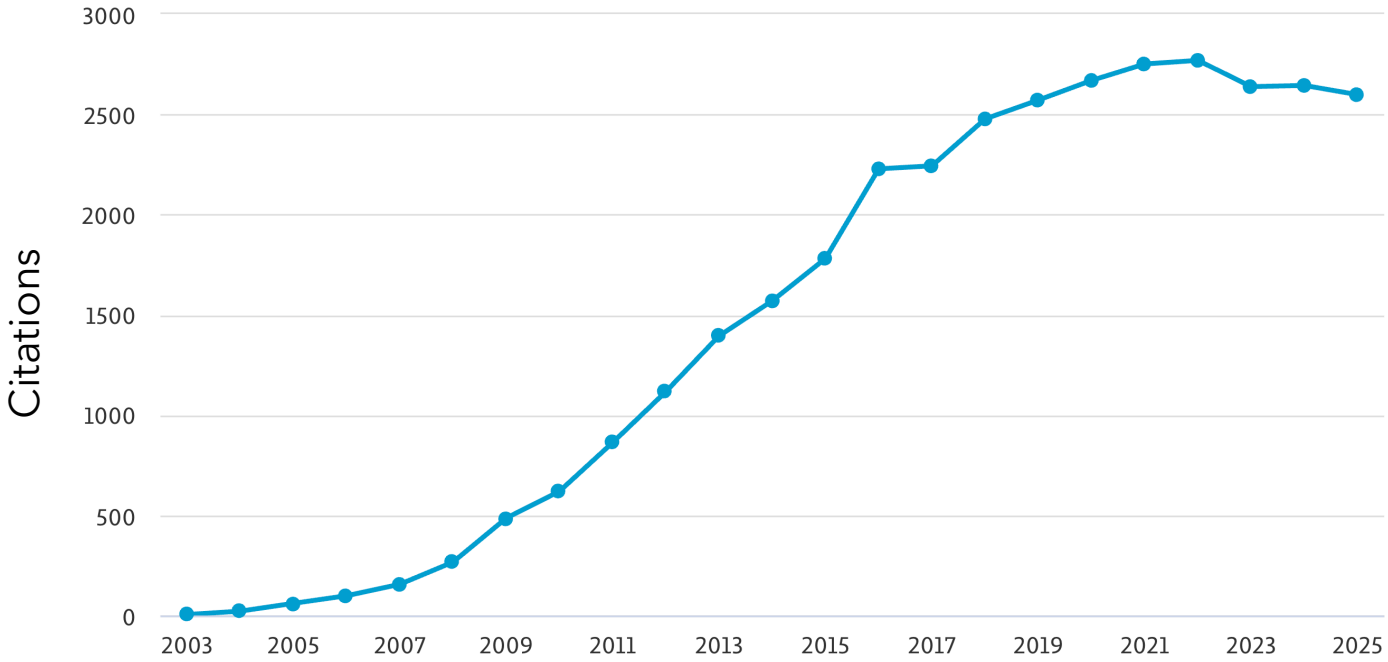


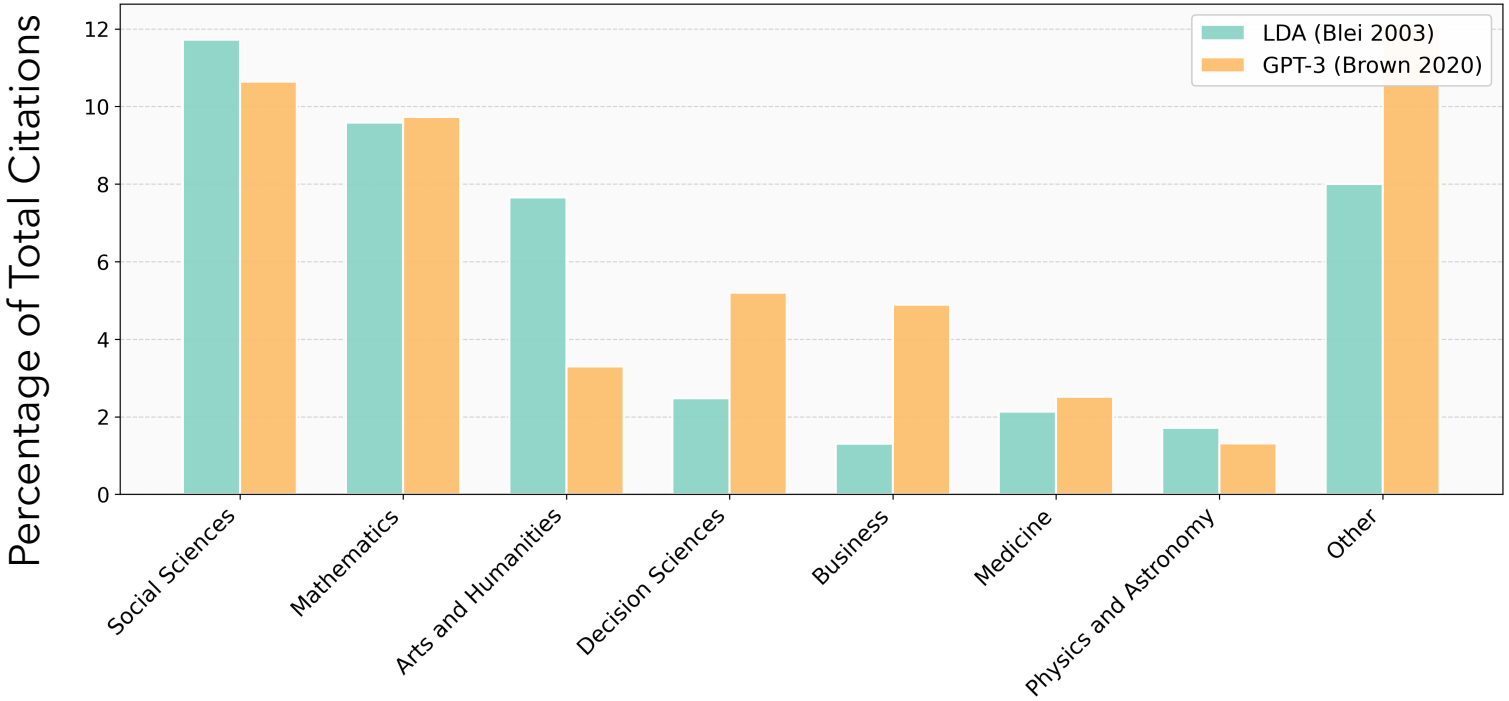

Text Corpus
Topic Model
Corpus-Level View
Document-Level View

Why talk about topic models?

- **Robust:** Over two decades of continuous use
- **Useful:** Large uptake outside machine learning
- **Interpretable:** White-box, Bayesian graphical model
- **Efficient:** Runs on a laptop or in a browser
- **Implementation-agnostic**: A testbed for different estimation methods
- **Pedagogical:** Illustrative of key concepts in ML,



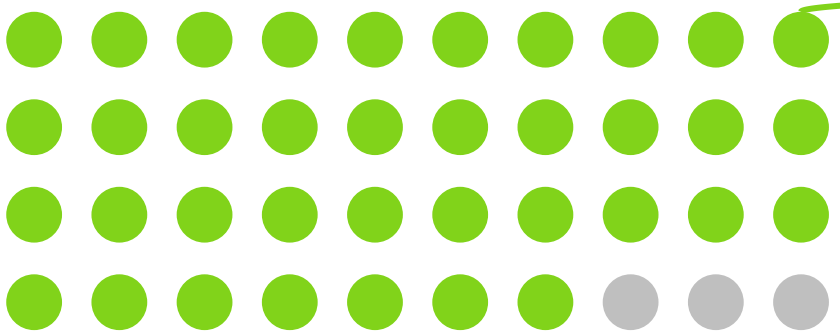
Topic Model (LDA) Citations by Year



Citation Distributions by Discipline (w/o CS & Engineering)

What do we use topic models for?

Mini meta-analysis of fifty non-CS papers that cite LDA (Blei et al. 2003).

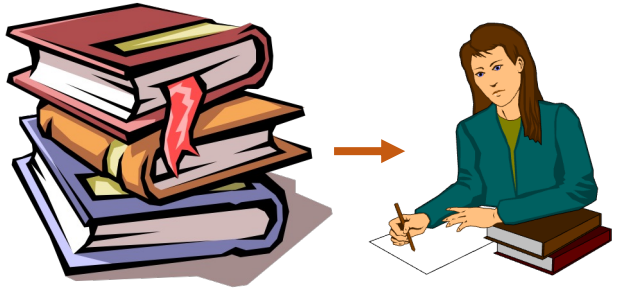


In 94% of papers, topic models are used for the **inductive discovery of categories** intended for human consumption

Hoyle et al. 2022. *Are Neural Topic Models Broken?* Findings of EMNLP.

A simplified picture of qualitative content analysis

Inductive codeset discovery

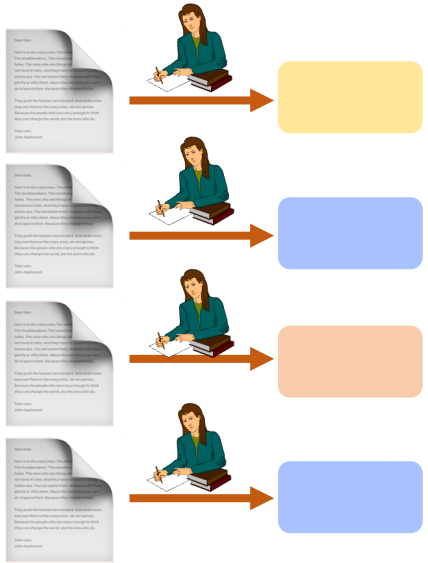


Books about art

Books about literature

Books about clothing

Deductive coding



Stemler, Steve (2000). “An overview of content analysis”
Smith, Charles P (2000). “Content analysis and narrative analysis”.

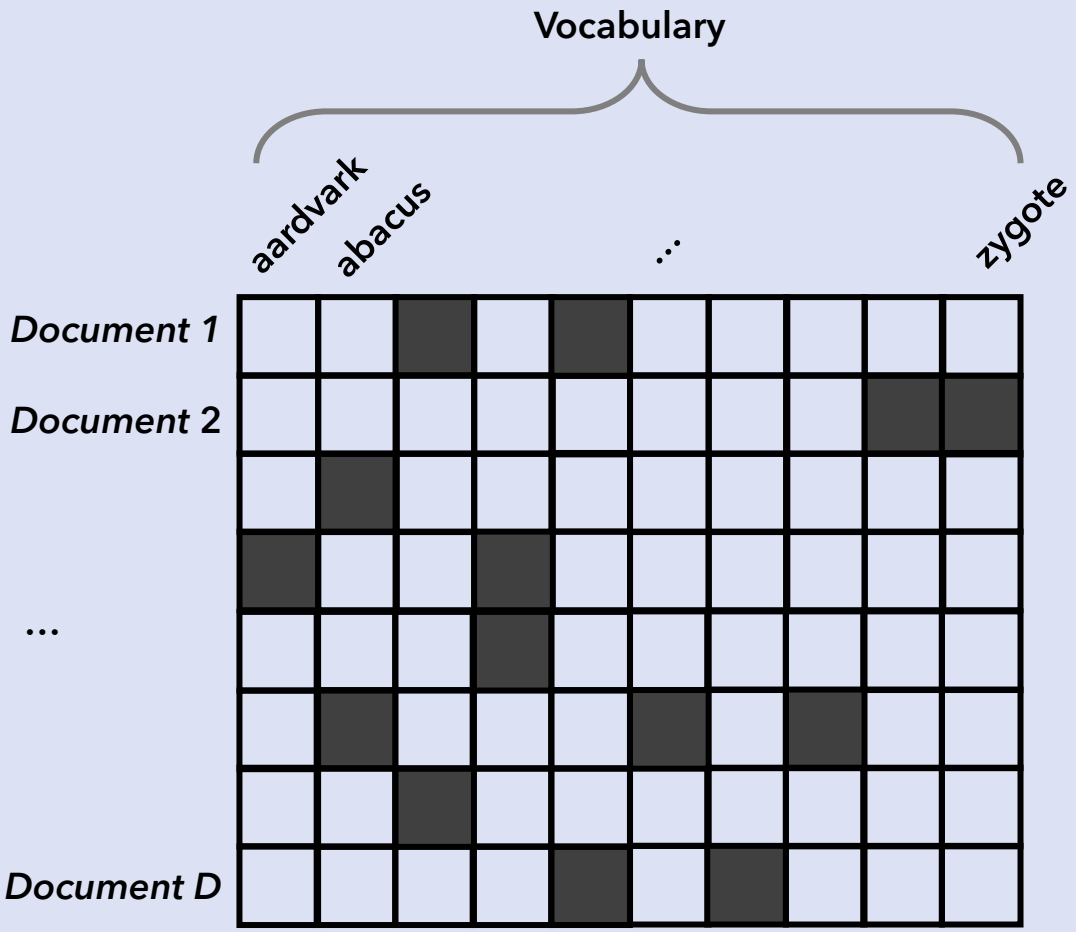
Topic models can help content analysis

Can be done manually, following a structured method like ***qualitative content analysis***

Can be facilitated by various NLP methods, like topic models

Understanding topic models

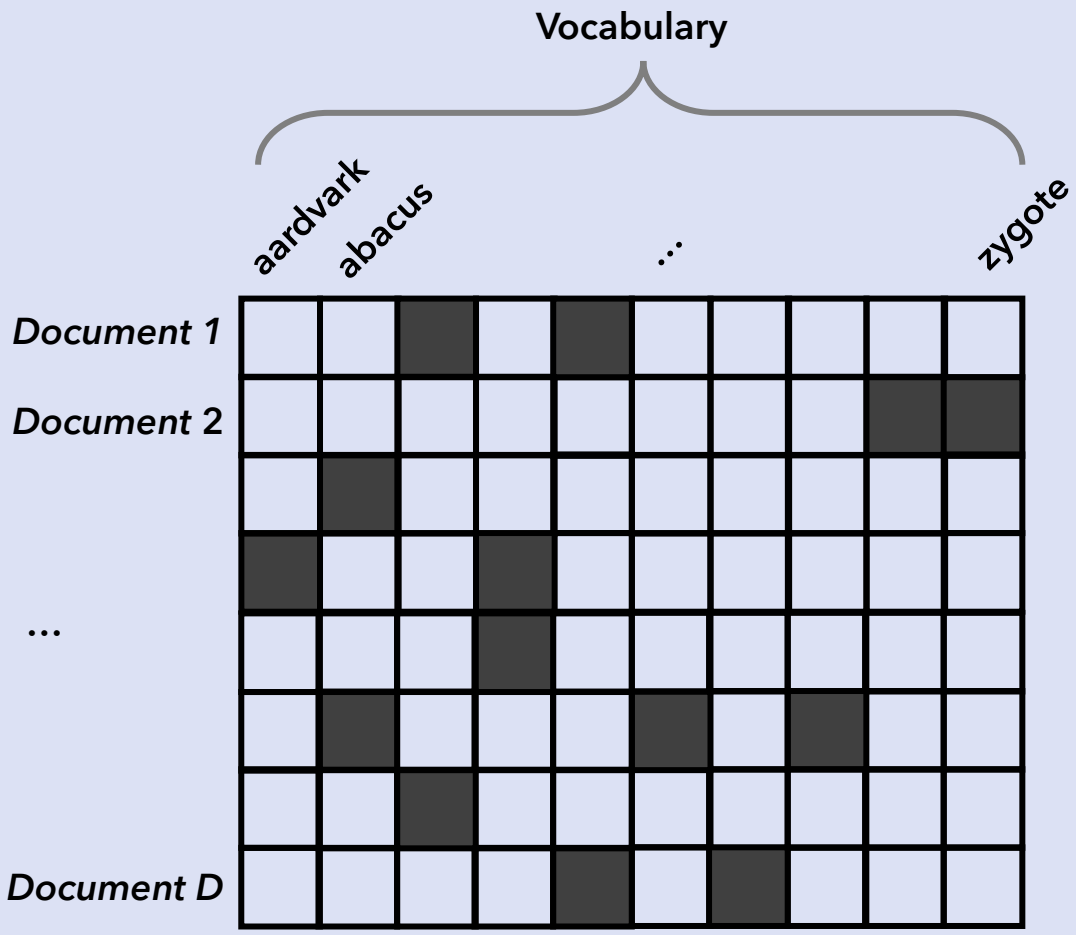
Represent documents as a bag-of-words



Sparse document-word matrix

$[D \times V]$

Observed Data

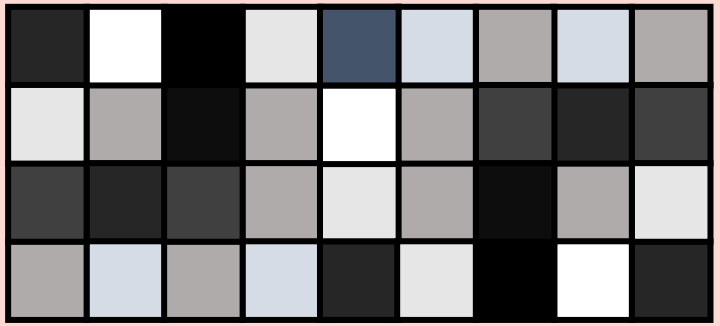


Sparse document-word matrix

$[D \times V]$

Latent Parameters

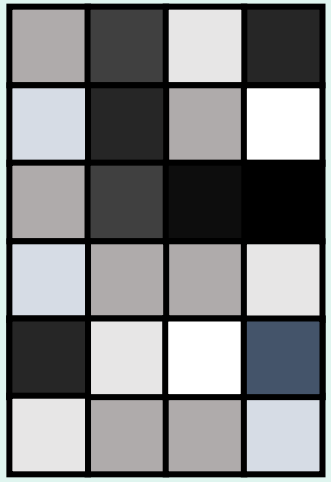
$\mathbf{B}$



Topic-word distributions

$[K \times V]$

$\boldsymbol{\Theta}$

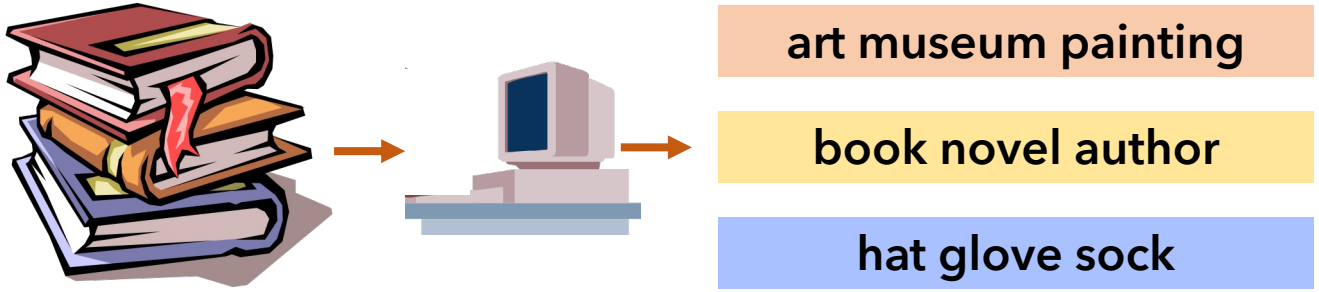


Document-topic distributions

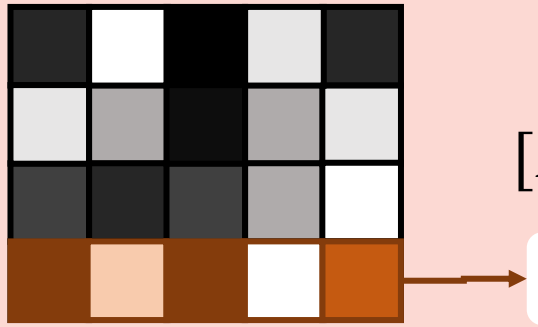
$[K \times V]$

Topic models as non-negative Bayesian matrix factorization

Estimated topic-word distribution $\mathbf{B}_{k,\cdot}$



$\mathbf{B}$

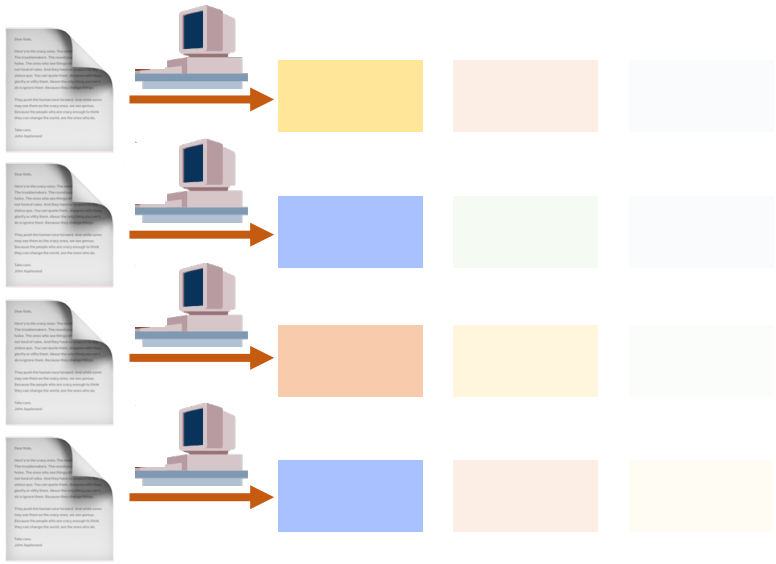


$[K \times V]$ Global topic-word distributions

head hair brain frog hat

Topic models as non-negative Bayesian matrix factorization

Estimated document-topic distribution $\boldsymbol{\Theta}_{\cdot,\boldsymbol{k}}$

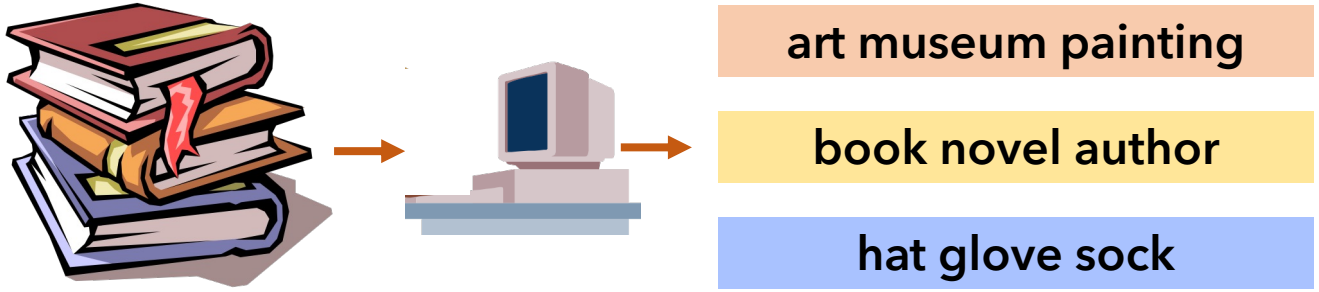


$\boldsymbol{\theta}_d$

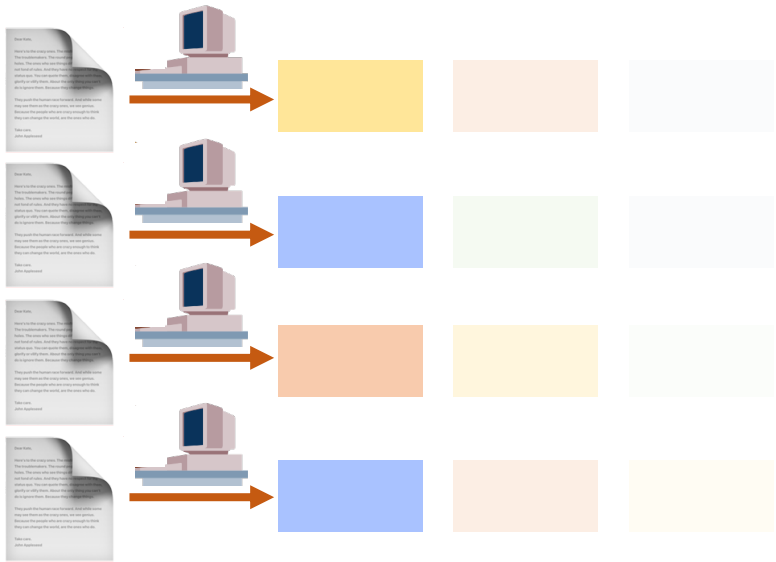
$[1 \times K]$ **A document d's distribution over K topics**

Topic models align with qualitative content analysis

Estimated topic-word distribution $\mathbf{B}_{k,\cdot}$



Estimated document-topic distribution $\mathbf{\Theta}_{\cdot,k}$



Topic models help understand a corpus

vote democracy voting right rights elections people

judge supreme_court senate justice nomination

energy clean jobs water gas oil climate emissions air

debt spending democrats bill student trillion tax

border crisis biden immigration president southern

Topic models are non-negative Bayesian matrix factorization

For each document d:

Draw its distribution over K topics

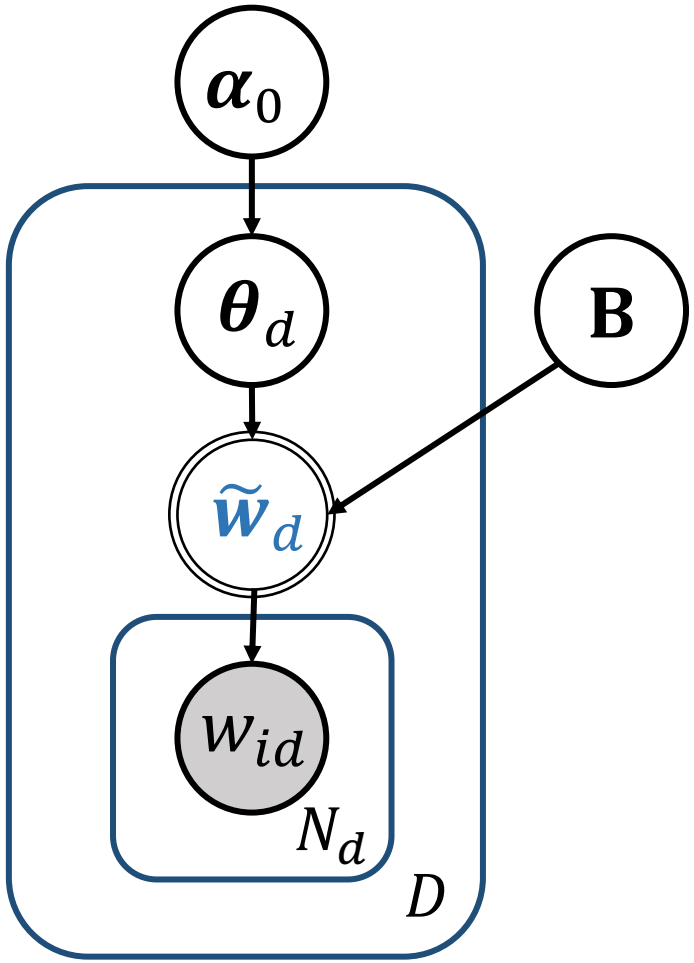
$\boldsymbol{\theta}_d \sim \text{Dirichlet}(\alpha_0)$

Calculate word distribution parameters

$\widetilde{\boldsymbol{w}}_d = softmax(\boldsymbol{\theta}_d^{\mathrm{T}}\mathbf{B})$

For each of the N_d words w_{id} in the document:

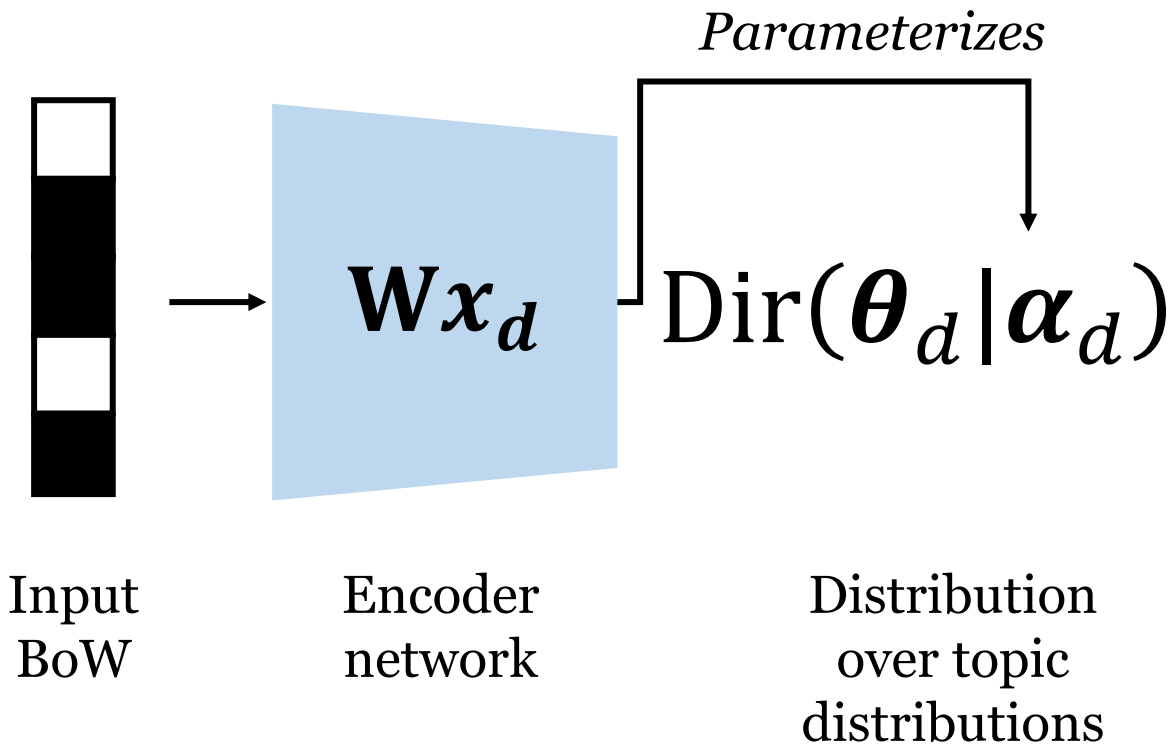
$$w_{id} \sim \text{Multinomial}(1, \widetilde{\boldsymbol{w}}_d)$$



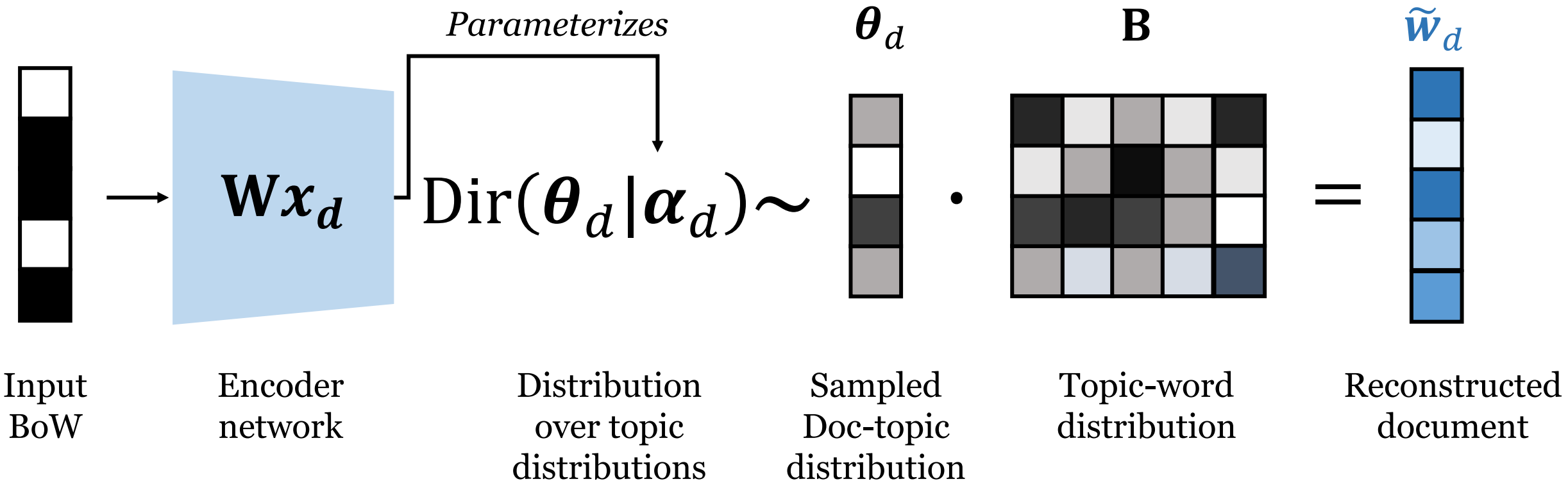
Inference is not so easy

$$p(\boldsymbol{\theta}_d \mid \mathbf{w}_d, \mathbf{B}) = \frac{p(\mathbf{w}_d \mid \boldsymbol{\theta}_d, \mathbf{B})\, p(\boldsymbol{\theta}_d)}{p(\mathbf{w}_d)}$$

Encoder parameterizes a document's topic distribution



Decoder generates a reconstruction of the document



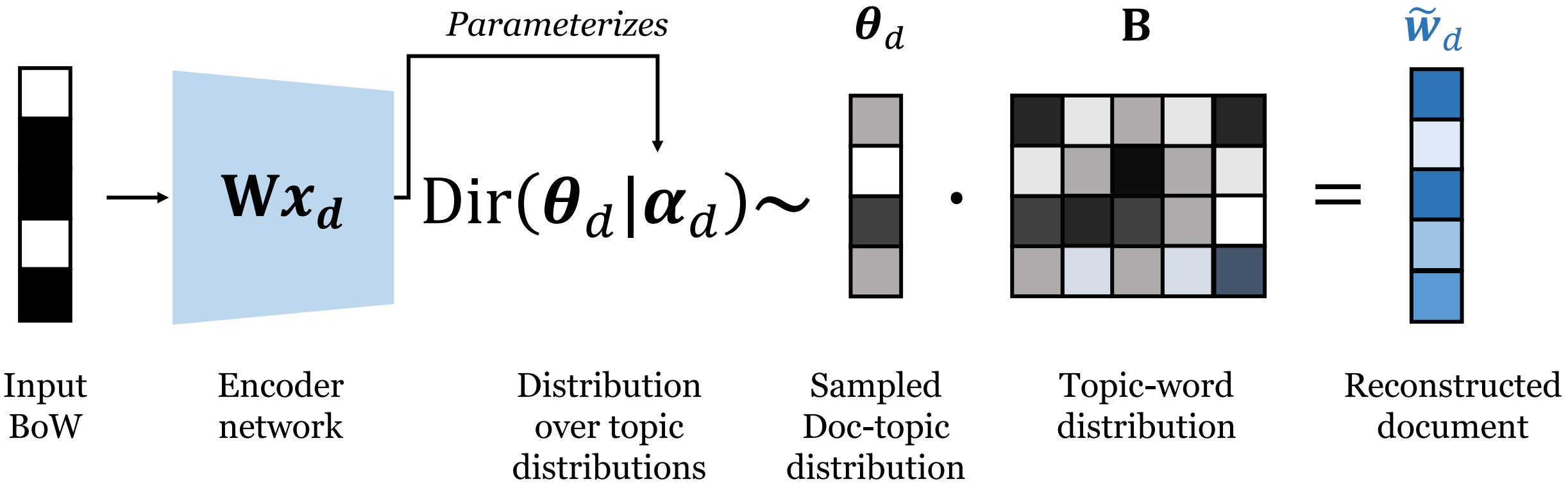
Loss: expected reconstruction error and KL divergence

$$\tilde{\boldsymbol{w}}_d = softmax(\boldsymbol{\theta}_d^{\mathrm{T}}\mathbf{B})$$

$$-\mathrm{ELBO} = -\mathbb{E}_{q(\theta_d|\cdot)}\left[\boldsymbol{w}_{\mathrm{d}}^{\mathrm{BoW}^{\mathsf{T}}} \log \tilde{\boldsymbol{w}}_d\right] \quad \Big\} \text{ Reconstruction error}$$

$$+\mathrm{KL}\left[q\left(\theta_d \middle| \boldsymbol{w}_{\mathrm{d}}^{\mathrm{BoW}}, x_d\right) \parallel p(\theta_d)\right] \quad \Big\} \text{ KL between prior and approximate posterior}$$

Much flexibility in design choices



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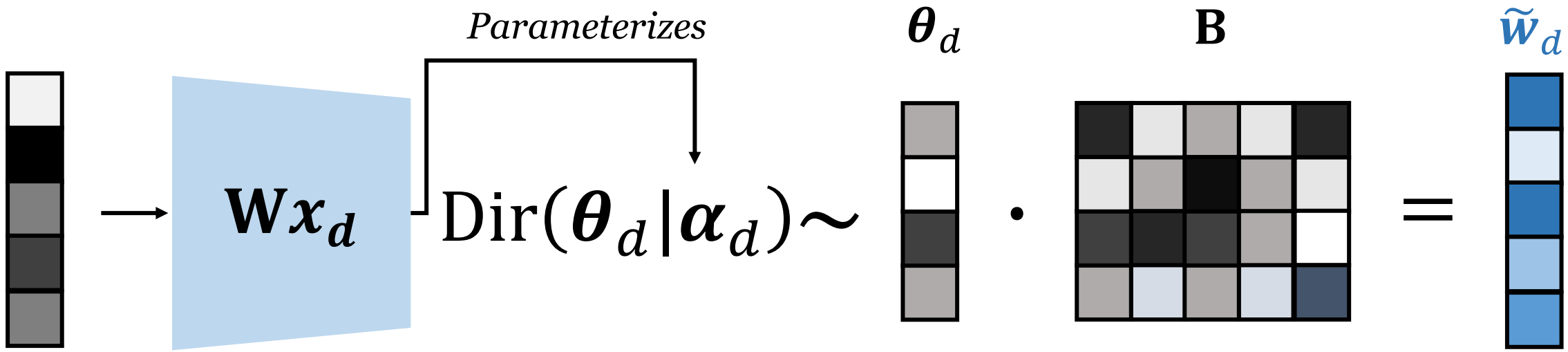


Sentence Embedding

Encoder network

Distribution over topic distributions

Sampled Doc-topic distribution

Topic-word distribution

Reconstructed document

Much flexibility in design choices

WHAI: WEIBULL HYBRID AUTOENCODING INFERENCE FOR DEEP TOPIC MODELING

Topic Modeling in Embedding Spaces

Contrastive Learning for Neural Topic Model

NEURAL TOPIC MODEL VIA OPTIMAL TRANSPORT

Improving Neural Topic Models using Knowledge Distillation

Topic Modeling with Wasserstein Autoencoders

Effective Neural Topic Modeling with Embedding Clustering Regularization

TOPIC MODELING AS MULTI-OBJECTIVE CONTRASTIVE OPTIMIZATION

How to evaluate? Well, it is a model of language...

From the encoder

$$\mathbf{w}_d^{\mathrm{BoW}^\top} \log \tilde{\mathbf{w}}_d = \log p(\mathbf{w}_d \mid \boldsymbol{\theta}_d, \mathbf{B})$$

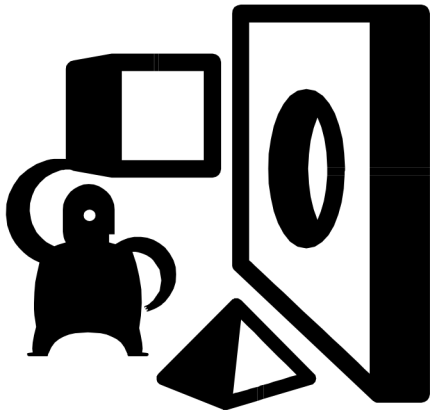
Reconstruction error is language model loss

$$\mathcal{L}_{\mathrm{recon}} = \sum_{d \in \mathcal{D}^{\mathrm{test}}} \log p(\mathbf{w}_d \mid \boldsymbol{\theta}_d, \mathbf{B})$$

$$\mathrm{perplexity} = \exp\left(-\frac{1}{N} \sum_{d \in \mathcal{D}^{\mathrm{test}}} \log p(\mathbf{w}_d \mid \boldsymbol{\theta}_d, \mathbf{B}) \right)$$

Vessels for ideas

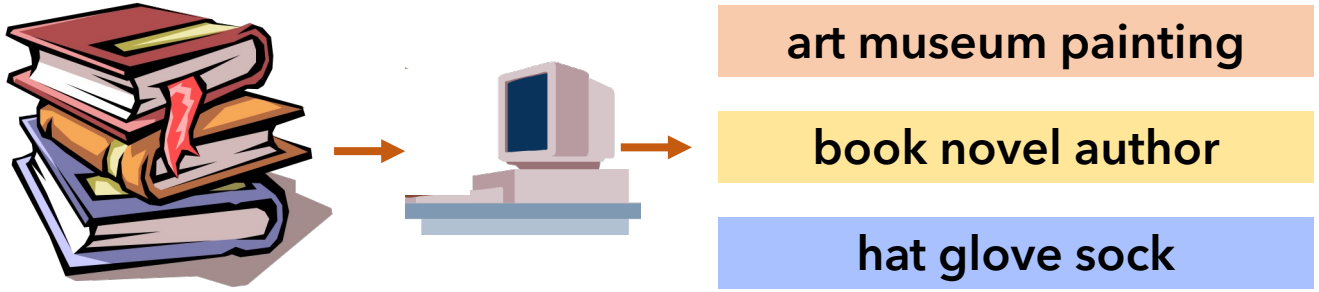
Tools to solve problems



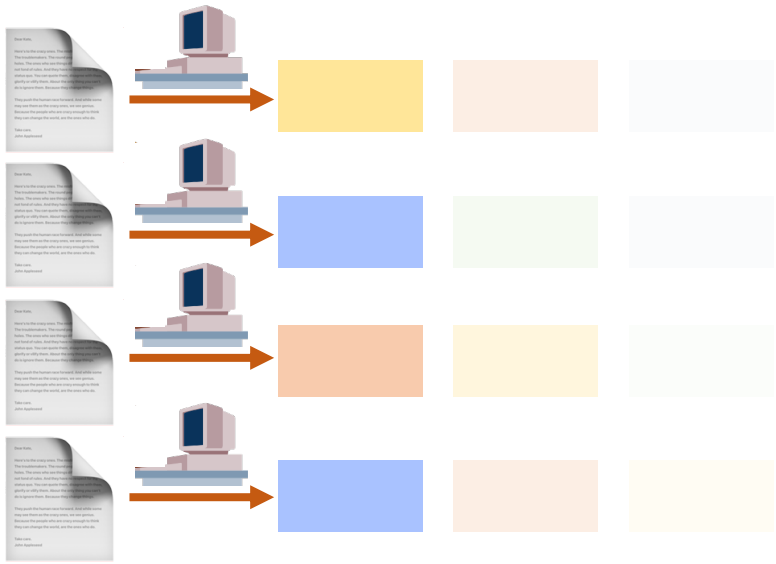
Estimating latent variables that minimize error is **not** the same as inferring *useful* ones

Measuring what we care about

Estimated topic-word distribution $\mathbf{B}_{k,\cdot}$



Estimated document-topic distribution $\mathbf{\Theta}_{\cdot,k}$



Are these any good?

What do topic model estimates look like?

Domestic environmentalism

$\mathbf{B}_{k,\cdot}$ public lands protect national parks state act

$\mathbf{\Theta}_{\cdot,k}$

▼ Documents

Great news for conservation! My bipartisan DESCEND Act will soon be signed into law. As a fisherman, I know how important sustaining a healthy reef fish population is for the sport & the economy. This new law will also ensure future generations can enjoy red snapper season! With the DESCEND Act I introduced w/@SenBillCassidy this week, we're aiming to protect our reef fish and ensure future generations can enjoy this pastime, too.

The greater #Chaco region is a #NM treasure that holds spiritual, cultural & historical significance to many Tribes & Pueblos. Proud to introduce a bill w/ @MartinHeinrich to protect this sacred archaeological wonder from expanding energy development & preserve this special place

.@SecBernhardt just announced the opening & expansion of more than 850 hunting & fishing opportunities across 2.3 million acres at 147 national wildlife refuges & national fish hatcheries. This final rule is great news to hunters & fishermen in WY & across the country

Today I visited the @HopewellOhioNPS & was able to see firsthand the unmet maintenance needs, including the exterior of the visitor center & trail repair. My Restore Our Parks Act will benefit this park as well as the 7 other national park sites in #Ohio

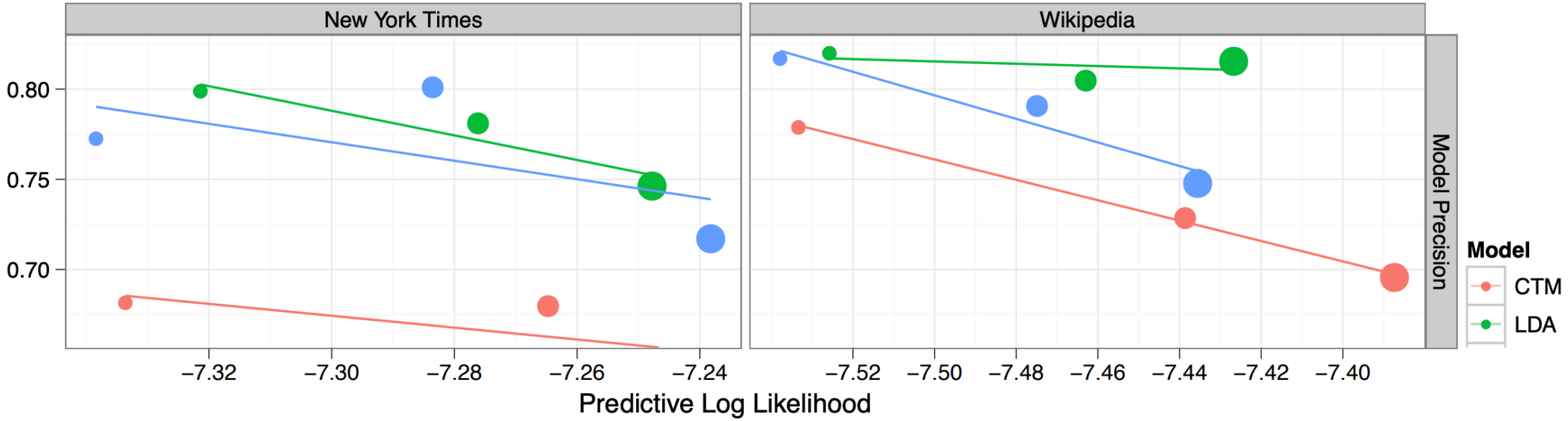
Topic model evaluation relies on topic-word **coherence**

$\mathbf{B}_{k,\cdot}$ public lands protect motherboard parks state act

Which word does not belong?

Chang et al. 2009. *Reading Tea Leaves: How Humans Interpret Topic Models*. NeurIPS.

Predictive log-likelihood is *negatively correlated* with a measure of topic coherence



Chang et al. 2009. *Reading Tea Leaves: How Humans Interpret Topic Models*. NeurIPS.

Topic model evaluation relies on topic-word coherence

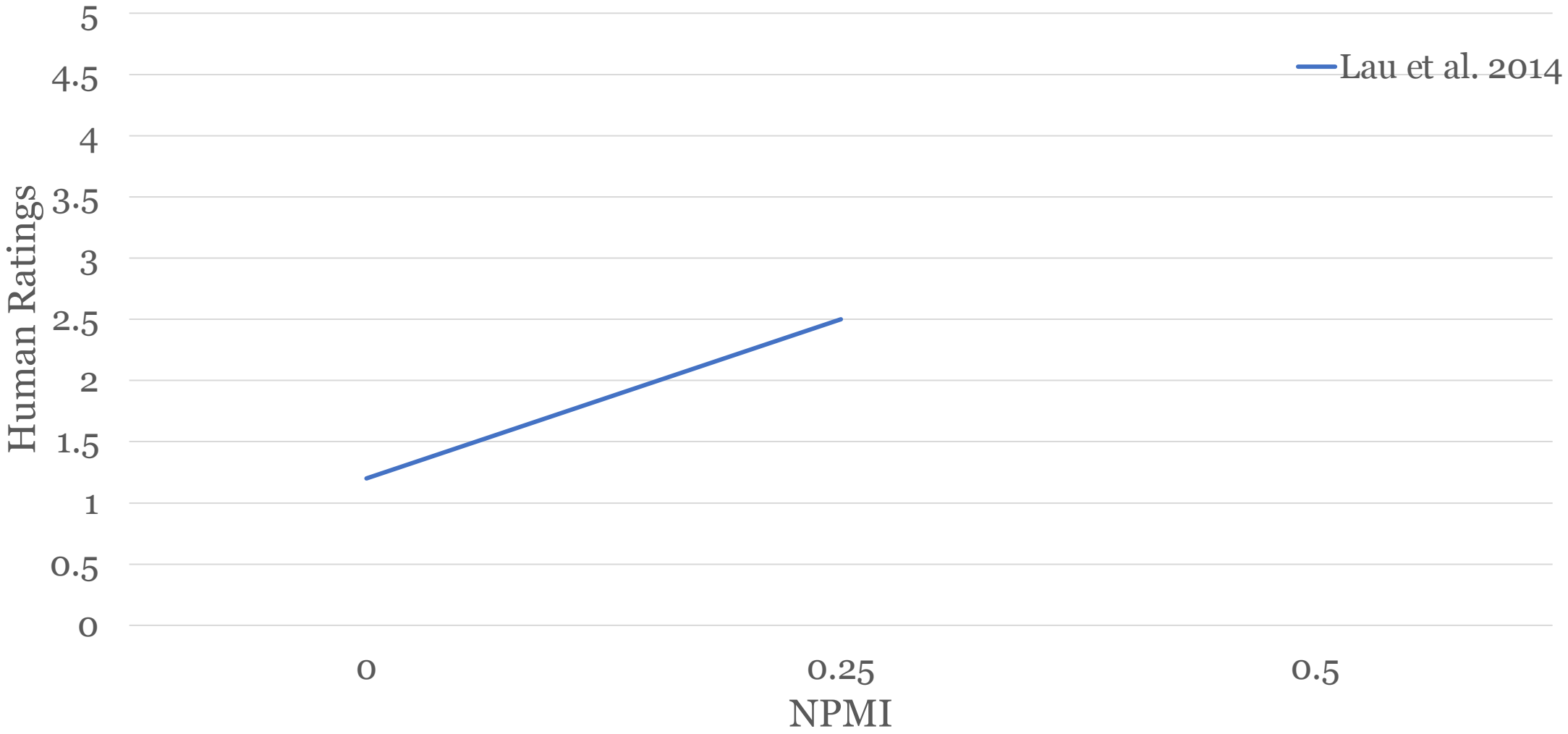
$$\sum_{j=2}^{N}\sum_{i=1}^{j-1}\frac{log\frac{P(w_j,w_i)}{P(w_i)P(w_j)}}{-logP(w_i,w_j)}$$

$\mathbf{B}_{k,\cdot}$ public lands protect motherboard parks state act

Normalized Pointwise Mutual Information (NPMI)

0.08

Why NPMI? It correlates with human ratings



Lau et al. 2014. *"Machine Reading Tea Leaves: Automatically Evaluating Topic Coherence and Topic Model Quality."* EACL

Topic model papers use automated coherence to compare the proposed model against baselines

- Meta-analysis of **40** newer, neural topic modeling papers:
  - Some form of automated coherence is used in **36 cases**

Hoyle et al. 2021. *Is Automatic Topic Model Evaluation Broken?* NeurIPS

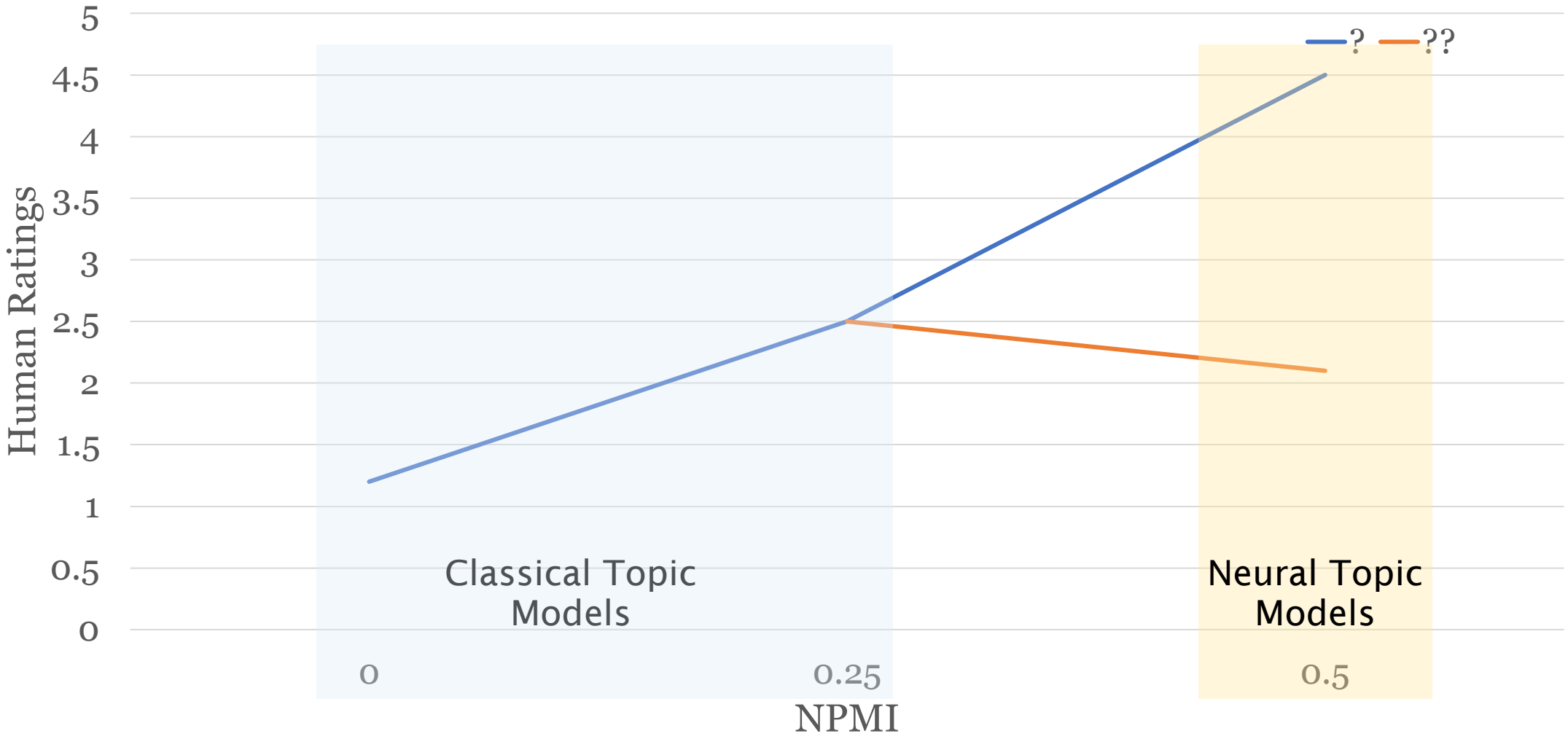
Topic model papers use automated coherence to compare the proposed model against baselines

- Meta-analysis of **40** newer, neural topic modeling papers:
  - Some form of automated coherence is used in **36 cases**
  - *But…*

Zero human evaluations!

Hoyle et al. 2021. *Is Automatic Topic Model Evaluation Broken?* NeurIPS

Can we be sure human correlations will hold?



Experimental setup

3 topic models

LDA (Mallet)

Dirichlet-VAE

Embedded TM

2 datasets

Wikipedia

New York Times

3 evaluations

$$\sum_{j=2}^{N}\sum_{i=1}^{j-1}\frac{log\frac{P(w_j,w_i)}{P(w_i)P(w_j)}}{-logP(w_i,w_j)}$$

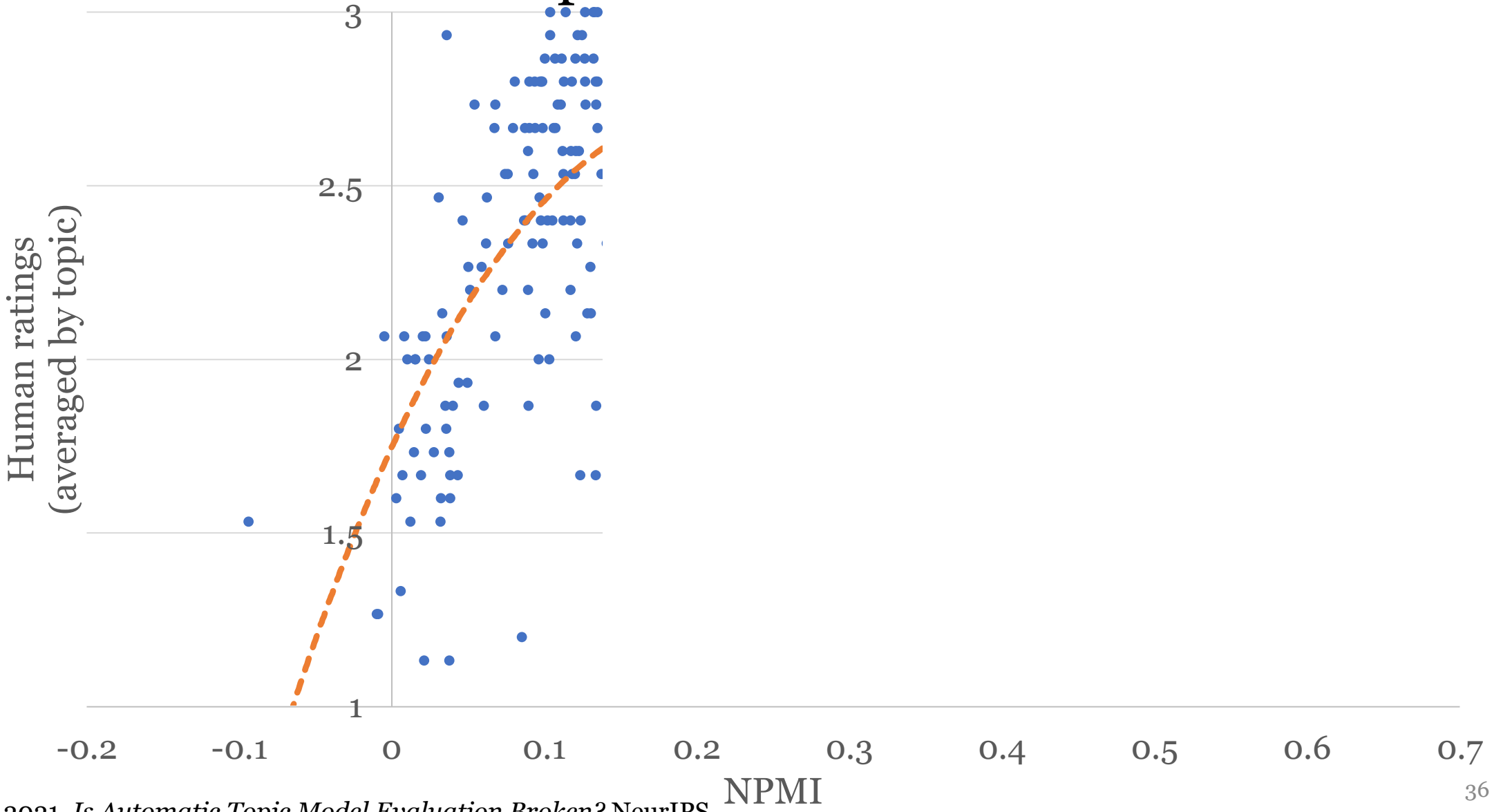
Automated coherence

Human ratings

`apple, plum, mainframe, peach`

Human word intrusion

After a certain point, Automated Coherence looses connection to human preferences



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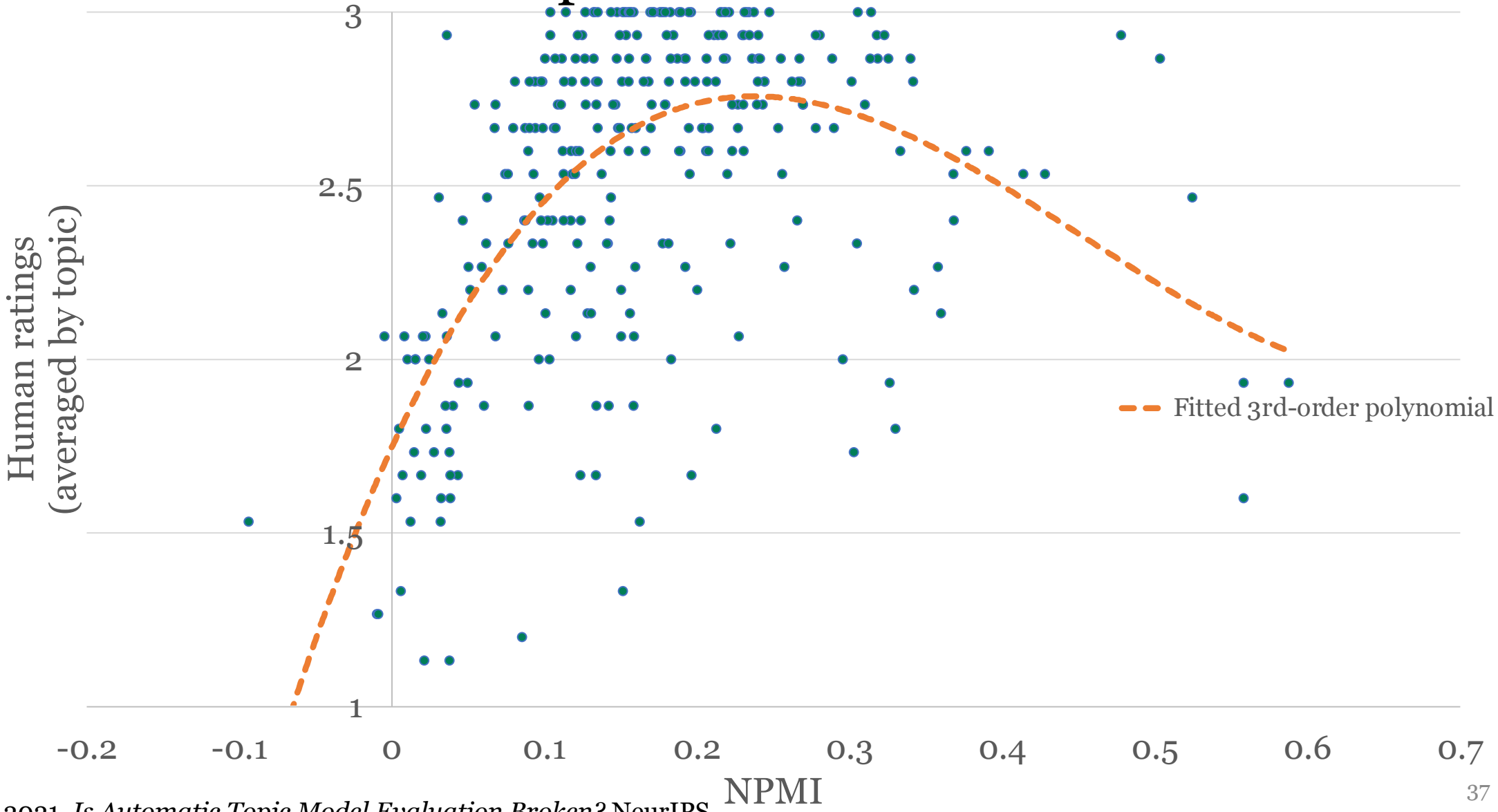

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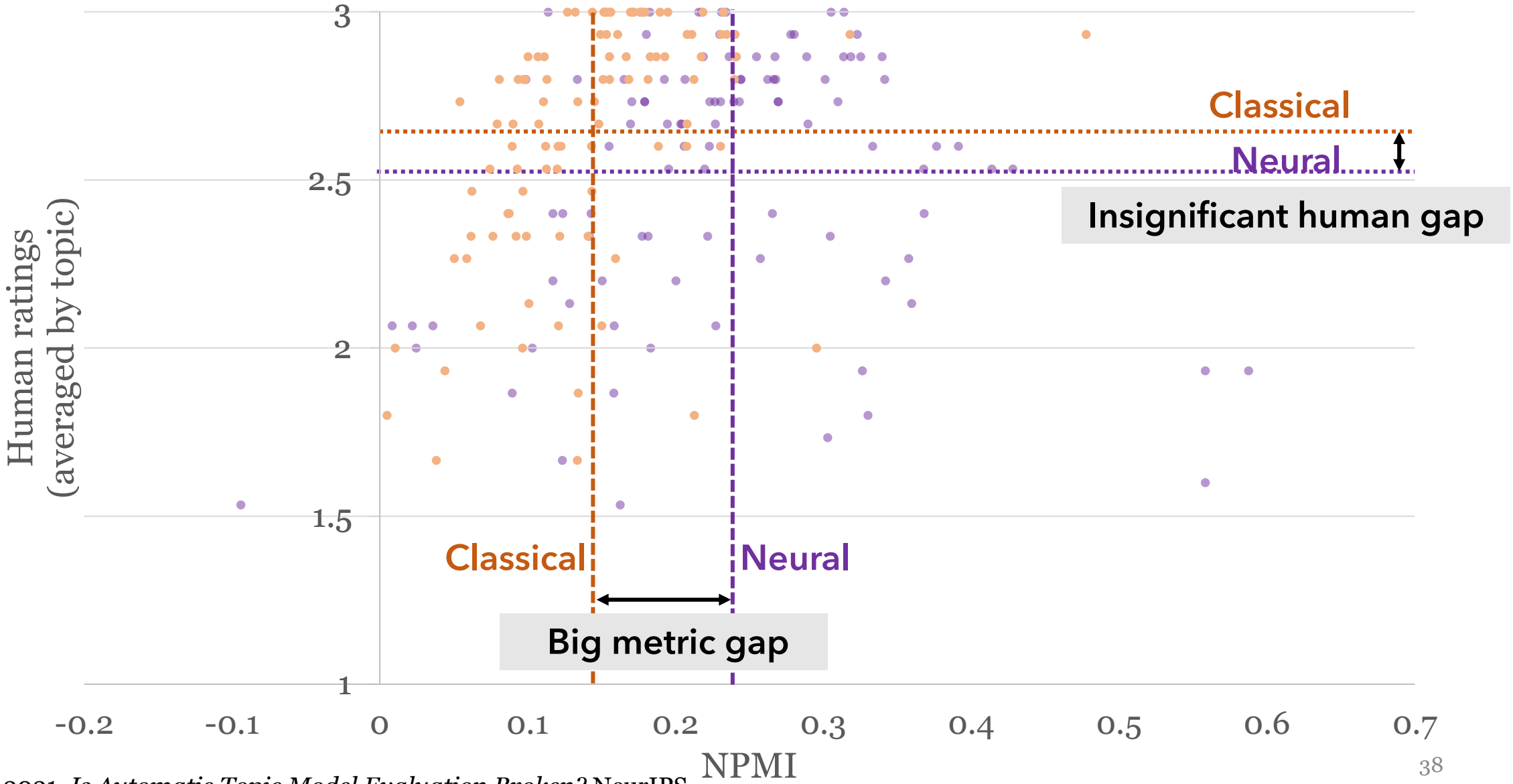
D-VAE has higher NPMI, but similar ratings to LDA

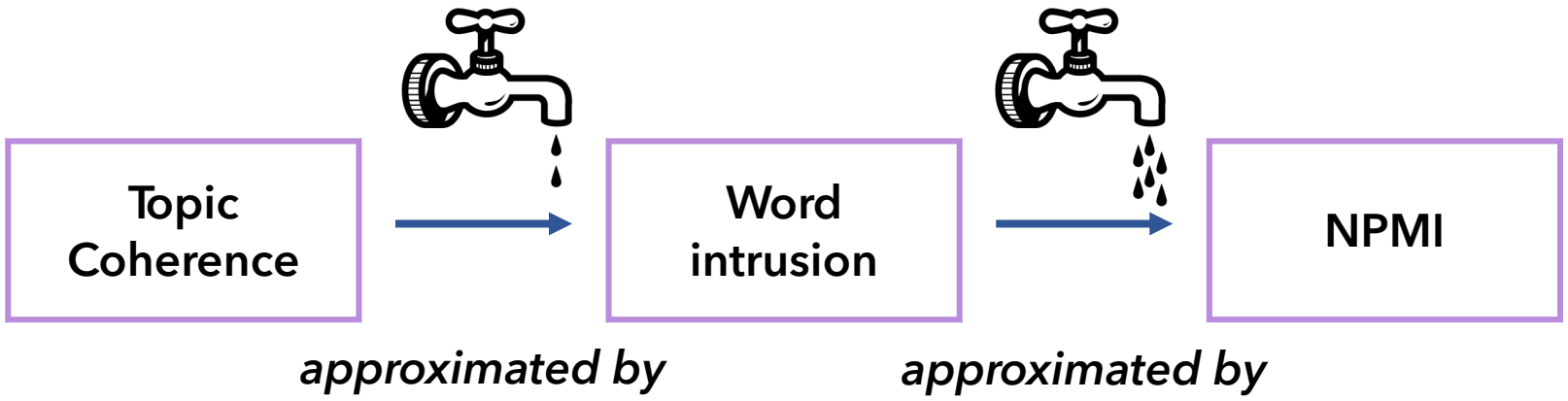


Hoyle et al. 2021. *Is Automatic Topic Model Evaluation Broken?* NeurIPS

So…how to evaluate topic models?

Hoyle et al. 2025 ProxAnn: Use-Oriented Evaluations of Topic Models. ACL 2025

Is coherence worth chasing?



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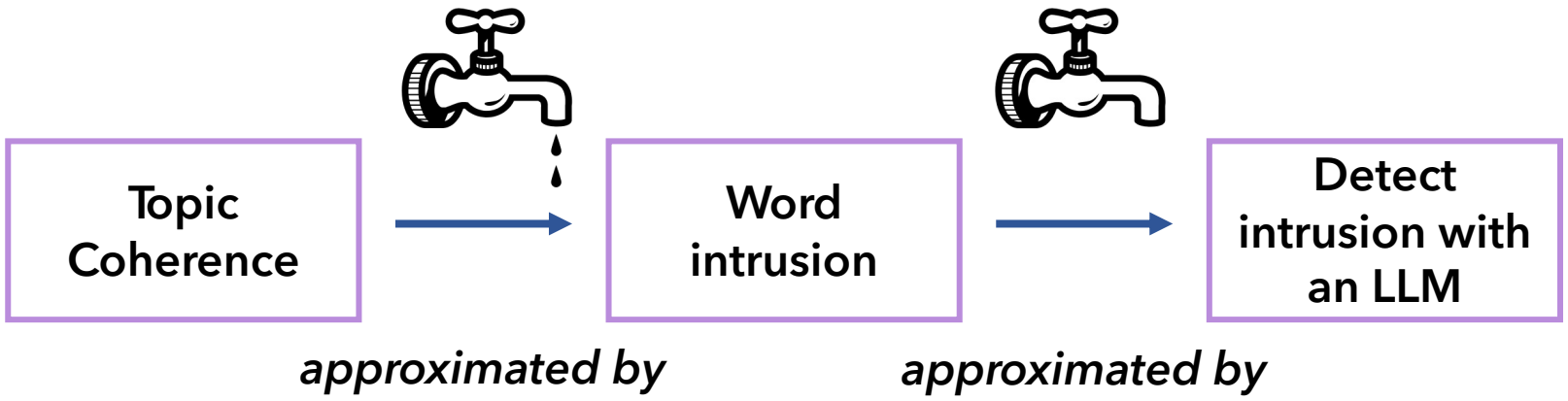

Stammbach et al. 2023. *Revisiting Automated Topic Model Evaluation with Large Language Models*. EMNLP

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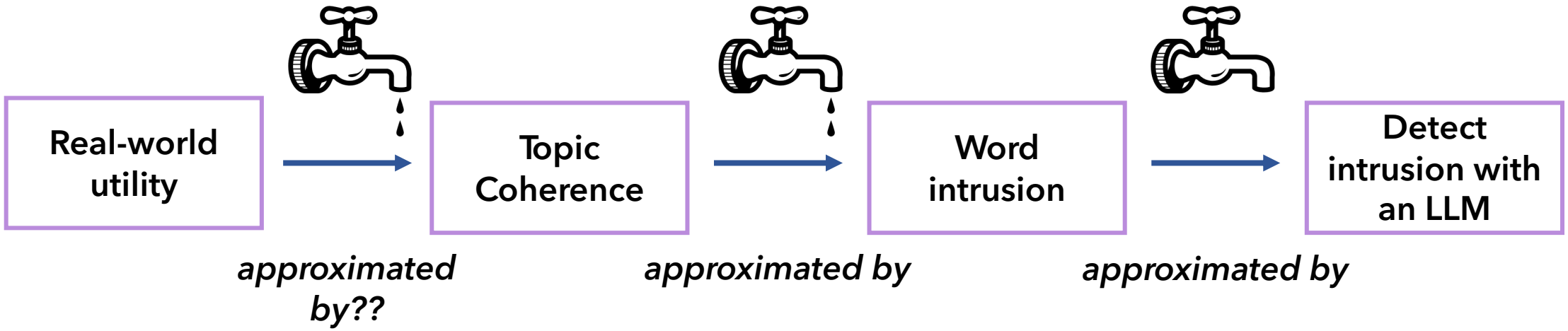
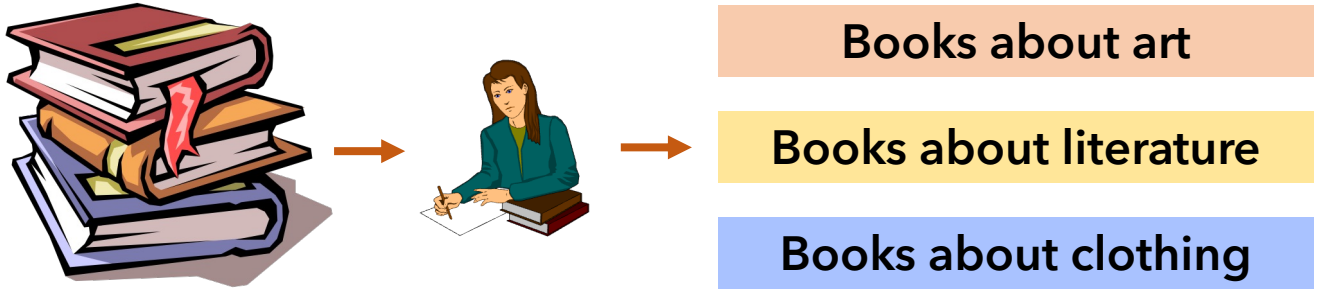


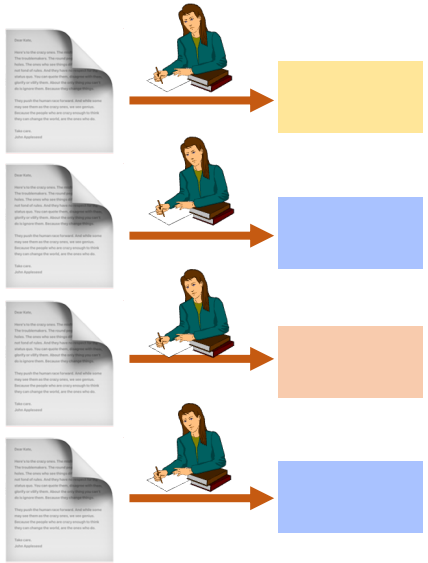
Stammbach et al. 2023. *Revisiting Automated Topic Model Evaluation with Large Language Models*. EMNLP

Idea: an evaluation protocol based on content analysis

Inductive codeset discovery



Deductive coding



LLM topic scores correlate well with human scores

Parting thoughts

- Evaluations should be validated: measure what you intend to measure
  - Adcock and Collier. 2001. *Measurement Validity: A Shared Standard for Qualitative and Quantitative Research*. APSR
  - Jacobs and Wallach. 2021. *Measurement and Fairness*. FAccT
  - Liao and Xiao. 2023. *Rethinking Model Evaluation as Narrowing the Socio-Technical Gap*
  - Wallach et al. 2025. *Evaluating Generative AI Systems Is a Social Science Measurement Challenge*. ICML
- Topic models are useful, if used right (forthcoming paper)
  - LDA estimated with Gibbs sampling is still highly effective (use Tomotopy or Mallet implementations)
  - LDA with variational inference gives subpar results (like Gensim)
  - BERTopic give substandard results. Avoid it.
  - LLM labels of topics hallucinate with nonnegligible frequency